SIMATS ENGINEERING

ASSESSMENT TOOL 2: Industry Problem Solving Task

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

COURSE OUTCOME COVERED

CO4: *Analyze big data characteristics and challenges across domains including security and application aspects.* (BL4)

Dave's Taxonomy: Articulation (Level 4) **Total Marks: 40**
Question Count: 5
Duration :60 Minutes

Code of Conduct

I P. Guru/192521010 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student P. Guru

Assessment Tasks

Industry Problem Solving Task

1. **Retail Data Optimization Problem -** A retail chain operates both physical stores and an online platform. It generates large volumes of sales, inventory, and customer behavior data. The company wants real-time inventory tracking and demand forecasting. It also aims to deliver personalized marketing recommendations to customers. Design a big data architecture to support these requirements. Explain the tools, technologies, and analytics models you would use.

2. **Social Media Fraud Detection Problem -** A social media platform processes billions of user interactions daily. These include posts, comments, likes, and shares. The platform wants to detect fake accounts and misinformation quickly. Real-time or near real-time processing is required. Design a scalable big data solution for this use case. Explain the role of machine learning and stream processing frameworks.

3. **Government Data Platform Problem** A government agency is building a national data platform. It combines census, economic, and geospatial datasets. Different departments need access to shared and integrated data. The system must ensure interoperability across multiple agencies. Strict privacy and security requirements must be followed. Propose a governance and data management strategy.

4. **Banking Fraud Detection Problem**
   A bank processes millions of financial transactions every minute. It needs to detect fraudulent activities in real time. The system must analyze historical and streaming transaction data. Low latency and high accuracy are critical requirements. Design a big data pipeline for fraud detection. Explain the technologies and algorithms you would implement.

5. **Agriculture Smart Farming Problem**
   A smart farming system collects soil, weather, and crop data. Sensors and drones provide real-time field information. Farmers want insights to improve yield and resource usage. Data is generated

from multiple distributed sources. Design a big data solution for precision agriculture. Explain how analytics can support decision-making.

Industry Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Industry Context	20%	Demonstrates strong understanding of real-world problem context	Good understanding	Basic understanding	Weak understanding
Application of Big Data Concepts	25%	Applies highly relevant big data ideas and tools	Mostly appropriate application	Partial application	Weak application
Solution Design	25%	Solution is practical, scalable, and well-reasoned	Reasonable solution	Basic solution with gaps	Unclear solution
Security / Risk Awareness	15%	Strong awareness of security and operational risks	Reasonable awareness	Limited awareness	Poor awareness
Clarity and Professionalism	15%	Well-structured and professional response	Generally clear	Moderately clear	Poorly presented

Total Marks Awarded:

1.Retail Data Optimization Problem - A retail chain operates both physical stores and an online platform. It generates large volumes of sales, inventory, and customer behavior data. The company wants real-time inventory tracking and demand forecasting. It also aims to deliver personalized marketing recommendations to customers. Design a big data architecture to support these requirements. Explain the tools, technologies, and analytics models you would use.

ANS:

Retail Data Optimization Problem – Big Data Architecture

A modern retail chain operates through both physical stores and an online shopping platform. Every day, huge amounts of data are generated from point-of-sale systems, online transactions, inventory systems, customer searches, product reviews, mobile applications, and social media interactions. This data is generally characterized by high volume, velocity, variety, and complexity. Therefore, traditional databases alone may not be sufficient to process and analyze the data efficiently.

The retail organization requires a Big Data architecture that can support real-time inventory tracking, demand forecasting, customer behavior analysis, and personalized marketing recommendations. The proposed architecture should collect data from multiple sources, process both real-time and historical data, store it in scalable platforms, and apply machine learning models to generate useful business insights.

Proposed Big Data Architecture:

The architecture can be divided into the following major layers:

Data Sources → Data Ingestion → Stream/Batch Processing → Data Storage → Analytics and Machine Learning → Visualization and Applications

1. Data Sources:

The first layer consists of all systems that generate retail data. Physical stores generate transaction information through POS terminals, barcode scanners, billing systems, and inventory management systems. The online platform generates data from website searches, product views, shopping carts, orders, payments, and customer reviews.

Other sources include:

- Customer mobile applications
- RFID and IoT inventory sensors
- Supplier systems
- Social media platforms
- Loyalty programs
- Product databases
- Customer feedback and reviews

This creates structured, semi-structured, and unstructured data.

2. Data Ingestion Layer:

- The collected data must be transferred into the Big Data platform. Technologies such as Apache Kafka can be used for high-speed streaming data ingestion. Kafka can continuously receive events such as purchases, product searches, inventory updates, and customer activities.
- For batch data, tools such as Apache Sqoop or cloud-based data integration services can be used to transfer data from traditional relational databases into the Big Data environment.
- For example, when a customer purchases a product from a physical store, the transaction can immediately be published as an event to Kafka. The inventory system can then update the available stock.

3. Stream Processing:

- Real-time processing is essential for inventory tracking. Apache Spark Streaming or Apache Flink can process incoming Kafka streams.
- For example, suppose a particular mobile phone has only five units remaining. When a customer purchases one unit, the streaming system immediately updates the inventory count. If the quantity reaches a predefined threshold, the system can automatically generate a restocking alert.
- Stream processing can also identify unusual sales patterns, sudden demand increases, and product shortages.

4. Big Data Storage:

The retail organization needs scalable storage for both historical and real-time data.

Hadoop HDFS can be used for large-scale distributed storage. Alternatively, cloud object storage such as Amazon S3, Azure Data Lake Storage, or Google Cloud Storage can be used.

A data lake can store raw data in different formats, including:

- Transaction records
- JSON customer activity
- Product images
- Reviews
- Sensor readings
- Website logs

A structured data warehouse can then store cleaned and transformed data for reporting and business intelligence.

5. Data Processing:

Apache Spark can be used for large-scale batch processing and analytics. Spark can process millions or billions of historical transactions efficiently.

Data preprocessing involves:

- Removing duplicate records
- Handling missing values
- Correcting inconsistent product information
- Combining customer and transaction data
- Normalizing numerical values
- Creating useful analytical features

The processed data can then be used by machine learning algorithms.

6. Demand Forecasting:

Demand forecasting helps the retailer determine how many units of a product will be required in the future.

Historical sales data can be analyzed using models such as:

- Linear Regression
- Random Forest
- XGBoost
- ARIMA
- Prophet
- LSTM neural networks

For example, an LSTM model can analyze historical sales patterns, seasonal trends, holidays, promotions, and regional demand to predict future sales.

If the model predicts high demand for umbrellas during the rainy season, the company can increase inventory before demand rises.

7. Personalized Marketing Recommendations:

Customer behavior data can be analyzed to provide personalized product recommendations.

A collaborative filtering model can recommend products based on the behavior of similar customers. A content-based recommendation system can recommend products based on the customer's previous purchases and product preferences.

More advanced recommendation systems can use machine learning or deep learning to combine:

- Purchase history
- Search history
- Product views
- Customer preferences
- Demographic information
- Ratings and reviews

For example, if a customer frequently purchases sports shoes and fitness accessories, the system can recommend new sports products.

8. Data Visualization

Business users require dashboards to understand the information generated by the Big Data system. Tools such as Power BI, Tableau, or Grafana can display:

- Current inventory levels
- Sales performance
- Product demand
- Regional sales
- Customer behavior
- Forecasted demand
- Low-stock alerts

Managers can use these dashboards to make faster decisions.

9. Security and Governance

- Since customer and transaction data are sensitive, the system must include strong security mechanisms. Authentication, authorization, encryption, role-based access control, and data masking should be implemented.
- Data governance policies should define who can access customer information and how long the information should be retained.
- Working of the System
- When a customer purchases a product, the transaction is immediately sent to Kafka. The streaming engine processes the transaction and updates the inventory. At the same time, the transaction is stored in the data lake.

- Historical sales data is periodically processed using Spark. Machine learning models analyze historical and real-time data to forecast future demand. The recommendation engine analyzes customer behavior and generates personalized product suggestions.
- Finally, dashboards display the results to managers, while customers receive personalized recommendations through the website or mobile application.

Advantages:

The proposed architecture provides:

1. Real-time inventory visibility.
2. Accurate demand forecasting.
3. Reduced product shortages and overstocking.
4. Personalized customer experiences.
5. Faster business decision-making.
6. Scalable processing of large datasets.
7. Better supply-chain management.
8. Improved marketing effectiveness.

2. **Social Media Fraud Detection Problem - A social media platform processes billions of user interactions daily. These include posts, comments, likes, and shares. The platform wants to detect fake accounts and misinformation quickly. Real-time or near real-time processing is required. Design a scalable big data solution for this use case. Explain the role of machine learning and stream processing frameworks.**

   ANS:

   Social Media Fraud Detection – Scalable Big Data Solution
   Proposed Architecture
   The architecture can be represented as:
   Social Media Events → Kafka → Stream Processing → Feature Extraction → ML Models → Fraud/Misinformation Detection → Alerts and Action → Data Lake
   1. Data Sources:
   The system collects data from various social media activities, including:
   - Posts
   - Comments
   - Likes
   - Shares
   - Followers and following relationships
   - Login activities
   - IP/device information

- Account creation information
- Hashtags
- Images and videos
- User reports

This information contains both structured and unstructured data.

2. Data Ingestion

Apache Kafka can act as the primary data ingestion platform. Every user activity can be converted into an event.

For example:

User A → Likes → Post B

(or)

User C → Creates Account

These events are continuously published into Kafka topics.

Kafka is suitable because it provides high-throughput, fault-tolerant, distributed event processing.

3. Stream Processing

The incoming data must be analyzed quickly. Apache Flink, Spark Structured Streaming, or Kafka Streams can process the events.

Stream processing can calculate real-time indicators such as:

- Number of posts per minute
- Number of accounts created from one device
- Frequency of likes
- Sudden sharing patterns
- Repeated comments
- Unusual login locations
- Abnormally high follower growth

  For example, if an account creates hundreds of posts within a few minutes, the system can assign a high risk score.

4. Fake Account Detection:

Machine learning can be used to determine whether an account is genuine or fake.

Important features include:

- Account age
- Number of followers
- Number of following accounts
- Posting frequency
- Profile completeness
- Interaction ratio
- Device information
- Login behavior
- Network relationships

  Classification algorithms such as Random Forest, XGBoost, Logistic Regression, and neural networks can be trained using previously identified fake and genuine accounts.

  The model produces a probability or fraud score.

For example:
Fraud Score = 0.92
This means the account has a high probability of being fraudulent.

5. Graph-Based Fraud Detection:

- Social media data naturally forms a graph. Users can be represented as nodes, while follows, likes, comments, and shares can be represented as edges.
- Graph analytics can detect coordinated fraud networks.
- For example, if thousands of newly created accounts follow one another and repeatedly share the same content, the graph structure may indicate a bot network.
- Graph algorithms and graph neural networks can be used to detect these relationships**.**

6. Misinformation Detection:

Misinformation detection requires analysis of the actual content as well as user behavior.
Natural Language Processing (NLP) can analyze text using:

- Sentiment analysis
- Text classification
- Keyword analysis
- Transformer-based language models
- Similarity detection

A machine learning model can classify content as potentially reliable, suspicious, or misleading.
Images and videos can also be analyzed using computer vision models.

7. Real-Time Decision Engine

The results from multiple models can be combined into a risk score.
For example:

- Account behavior score = 80
- Content score = 70
- Network score = 90

The system can calculate an overall risk score.
If the score exceeds a predefined threshold, actions can include:

- Sending an alert
- Reducing content visibility
- Requesting account verification
- Temporarily restricting an account
- Sending content for human review

8. Data Storage:

- Historical events can be stored in a distributed data lake using Hadoop HDFS or cloud storage.

- Databases such as Cassandra, HBase, or cloud NoSQL databases can be used when high-speed access to large-scale user information is required.
- Historical data is important because it allows the machine learning models to be continuously retrained.

9. Machine Learning Model Training

- Historical labeled data can be used to train fraud detection models.

The process includes:

1. Collect historical interactions.
2. Label fraudulent and legitimate behavior.
3. Extract relevant features.
4. Train machine learning models.
5. Evaluate precision and recall.
6. Deploy the model.
7. Monitor model performance.
8. Retrain periodically.

   Since fraud patterns continuously change, the model should be updated regularly.

Scalability:

- The architecture should be distributed across multiple servers. Kafka partitions can distribute incoming events, while Spark or Flink can process streams in parallel.
- Cloud auto-scaling can increase computing resources when user activity increases.
- For example, during a major event, the platform may experience several times more interactions than normal. A distributed architecture can automatically scale to handle the additional load.
- Security and Privacy
- User information must be protected through encryption, access control, authentication, and secure APIs. Sensitive personal information should be minimized and protected according to applicable privacy regulations.

Advantages:

The proposed system provides:

1. Real-time fraud detection.
2. Early identification of fake accounts.
3. Rapid misinformation detection.
4. Scalable processing of billions of events.
5. Detection of coordinated bot networks.
6. Continuous machine learning improvement.
7. Reduced manual monitoring.
8. Improved platform trust and safety.

3. **Government Data Platform Problem A government agency is building a national data platform. It combines census, economic, and geospatial datasets. Different departments need access to shared and integrated data. The system must ensure interoperability across multiple agencies. Strict privacy and security requirements must be followed. Propose a governance and data management strategy.**

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4. Banking Fraud Detection Problem

A bank processes millions of financial transactions every minute. It needs to detect fraudulent activities in real time. The system must analyze historical and streaming transaction data. Low latency and high accuracy are critical requirements. Design a big data pipeline for fraud detection. Explain the technologies and algorithms you would implement.

ANS:

Banking Fraud Detection – Real-Time Big Data Pipeline

Proposed Architecture:

The architecture can be represented as:

Transaction Sources → Kafka → Stream Processing → Feature Engineering → Fraud Detection Models → Risk Engine → Alert/Block → Data Lake → Model Training

1. Transaction Sources

The system collects transaction information from:

- ATM transactions

- Credit card transactions
- Debit card transactions
- Internet banking
- Mobile banking
- UPI/payment systems
- POS terminals
- International transactions

Each transaction can contain information such as transaction amount, time, location, merchant, account ID, device information, and transaction type.

2. Data Ingestion

Apache Kafka can be used to receive transactions in real time.

Each transaction is published as an event. Kafka provides high throughput and fault tolerance and can distribute transactions across multiple partitions.

For example:

Transaction → Account A → ₹75,000 → Chennai → Online

This event can immediately enter the fraud detection pipeline.

3. Stream Processing

Apache Flink or Spark Structured Streaming can process transactions with low latency.

The stream-processing engine can calculate real-time features such as:

- Number of transactions in the last five minutes
- Total amount spent
- Average transaction amount
- Distance between consecutive transactions
- Number of failed login attempts
- New device usage
- Unusual merchant activity

For example, if a customer normally spends ₹2,000 per transaction but suddenly performs a ₹2,00,000 transaction from another location, the system can identify it as suspicious.

4. Rule-Based Detection

Rules provide fast initial fraud detection.

Examples include:

- Multiple transactions within a few seconds.
- Transactions from geographically impossible locations.
- Unusually high transaction amounts.
- Repeated failed authentication attempts.
- Transactions from blocked devices.
- Sudden changes in spending patterns.

Rules are useful because they are easy to understand and execute quickly.

5. Machine Learning

Machine learning provides more advanced fraud detection.

Classification models such as:

- Logistic Regression
- Random Forest
- XGBoost
- Gradient Boosting
- Neural Networks

can classify transactions as legitimate or fraudulent.
The model can use features such as transaction amount, location, time, merchant category, device type, and historical customer behavior.

6. Anomaly Detection

Fraudulent transactions may not always match previously known fraud patterns. Therefore, anomaly detection is important.
Algorithms such as:

- Isolation Forest
- One-Class SVM
- Autoencoders

can identify transactions that significantly differ from normal customer behavior.
For example, if a customer normally makes transactions within Chennai but suddenly performs several large transactions in multiple countries, the system may classify the behavior as anomalous.

7. Risk Scoring

The output from rules and machine learning models can be combined into a risk score.
For example:

- Rule score = 30
- Machine learning score = 45
- Anomaly score = 20

The combined score can determine the action.
Low risk: Approve automatically.
Medium risk: Request additional authentication.
High risk: Block or hold the transaction and alert the fraud investigation team.

8. Historical Data Storage

All transactions should be stored in a scalable data lake using cloud object storage or Hadoop-based storage.
Historical data is required for:

- Fraud investigation
- Customer behavior analysis
- Model training
- Regulatory reporting
- Pattern identification

A data warehouse can be used for business reporting and analytics.

9. Model Training

Historical transactions containing known fraud labels can be used to train machine learning models.

The training process involves:

1. Data collection.
2. Data cleaning.
3. Feature engineering.
4. Model training.
5. Model validation.
6. Performance measurement.
7. Deployment.
8. Continuous monitoring.

Because fraudulent transactions are usually much fewer than legitimate transactions, the dataset may be imbalanced. Techniques such as class weighting, oversampling, or suitable evaluation metrics can be used.

10. Low-Latency Processing

Fraud detection must occur within seconds or less. Therefore, the architecture should minimize unnecessary database queries and use in-memory processing where appropriate.

Stream processing allows the system to evaluate transactions as they arrive rather than waiting for batch processing.

11. Security

Banking systems require extremely strong security.

The platform should implement:

- Encryption
- Multi-factor authentication
- Role-based access control
- Secure APIs
- Network segmentation
- Audit logging
- Tokenization of sensitive information

Sensitive card and customer information should be protected throughout the pipeline.

Advantages:

The proposed system provides:

1. Real-time fraud detection.
2. Low-latency transaction analysis.
3. High scalability.
4. Better fraud detection accuracy.
5. Reduced financial losses.
6. Continuous learning from historical fraud.
7. Automated transaction risk scoring.
8. Strong security and auditability.

5. **Agriculture Smart Farming Problem**
A smart farming system collects soil, weather, and crop data. Sensors and drones provide real-time field information. Farmers want insights to improve yield and resource usage. Data is generated from multiple distributed sources. Design a big data solution for precision agriculture. Explain how analytics can support decision-making.

ANS:

Agriculture Smart Farming – Big Data Solution for Precision Agriculture

Proposed Architecture

The proposed architecture can be represented as:

Sensors/Drones/Satellites/Weather Data → IoT Gateway → Data Ingestion → Stream & Batch Processing → Data Lake → Analytics/ML → Farmer Dashboard and Automated Actions

1. Data Sources

The system collects information from multiple sources.

Soil sensors can measure:

- Soil moisture
- Temperature
- pH
- Nitrogen
- Phosphorus
- Potassium

Weather stations can provide:

- Temperature
- Humidity
- Rainfall
- Wind speed
- Solar radiation

Drones can capture high-resolution images of crops.

Satellite images can provide large-scale information about crop health and vegetation.

Other sources include GPS devices, irrigation systems, farm machinery, and historical agricultural records.

2. IoT Data Collection

Sensors continuously generate measurements. IoT gateways collect these readings and transmit them to the Big Data platform.

Protocols such as MQTT can be used for lightweight sensor communication.

For example, a soil sensor may send:

Field A → Moisture = 20%

If the moisture level falls below a predefined threshold, the system can generate an irrigation recommendation.

3. Data Ingestion

A streaming platform such as **Apache Kafka** can ingest large numbers of sensor events.

Since farms may contain hundreds or thousands of sensors, the architecture must support distributed data collection.

The incoming data can then be processed using **Apache Spark Streaming or Apache Flink**.

4. Data Storage

A data lake can store large amounts of agricultural information.

It can contain:

- Sensor readings
- Drone images

- Satellite images
- Weather records
- Crop information
- Soil information
- Historical yield data

Cloud object storage or Hadoop HDFS can be used for scalable storage.
Time-series databases can also be used for efficiently storing continuous sensor measurements.

5. Data Processing

Data preprocessing is necessary because sensor data may contain missing or incorrect values.
Processing activities include:

- Removing duplicate readings
- Handling missing values
- Removing sensor noise
- Normalizing measurements
- Combining sensor and weather information
- Linking data with GPS coordinates

The cleaned data can then be used for analytics.

6. Crop Yield Prediction

Machine learning models can predict crop yield using historical and current data.
Possible algorithms include:

- Linear Regression
- Random Forest
- XGBoost
- Artificial Neural Networks
- LSTM

Input variables can include:

- Soil moisture
- Soil nutrients
- Temperature
- Rainfall
- Crop type
- Previous yield
- Fertilizer usage

The model can predict expected yield for different sections of the farm.

7. Irrigation Optimization

One major advantage of smart farming is intelligent water management.
The system can combine soil moisture, weather forecasts, crop type, and historical water consumption to determine when irrigation is required.
For example, if soil moisture is already high and heavy rainfall is expected, the system can recommend delaying irrigation.
This reduces water wastage while maintaining crop health.

8. Crop Disease Detection

Drone and smartphone images can be analyzed using computer vision and deep learning.
Convolutional Neural Networks (CNNs) can identify visual symptoms of:

- Fungal infections
- Bacterial diseases
- Pest attacks

- Nutrient deficiencies

Early detection allows farmers to take corrective action before the disease spreads throughout the field.

9. Fertilizer Optimization:

Soil data can be analyzed to determine nutrient requirements.
Instead of applying the same quantity of fertilizer across the entire field, precision agriculture can recommend different quantities for different zones.
For example:
Zone A: High nitrogen → Low additional nitrogen requirement.
Zone B: Low nitrogen → Higher fertilizer requirement.
This reduces fertilizer costs and environmental impact.

10. Geospatial Analytics

GPS and GIS technologies can divide the farm into different management zones.
Each zone can be analyzed according to soil quality, crop health, moisture, and productivity.
Heat maps can show areas that require:

- More water
- Additional fertilizer
- Pest control
- Replanting
- Soil treatment

11. Farmer Dashboard

A mobile or web dashboard can present the results in a simple form.
The dashboard can display:

- Current soil moisture
- Weather conditions
- Crop health
- Predicted yield
- Irrigation recommendations
- Disease alerts
- Fertilizer recommendations

Farmers can therefore make decisions based on real-time information rather than relying only on manual observation.

12. Automated Decision Support

The system can also connect analytics results to farm equipment.
For example, if the system identifies low soil moisture in a particular zone, an automated irrigation system can activate only that zone.
This creates a closed-loop smart farming system:
Sense → Analyze → Decide → Act → Monitor

Advantages:

The proposed Big Data solution provides:

1. Increased crop yield.
2. Reduced water consumption.
3. Efficient fertilizer usage.
4. Early disease detection.
5. Better pest management.
6. Reduced farming costs.
7. Real-time monitoring.

8. Better resource utilization.
9. Data-driven decision-making.
10. Improved sustainability.